



Remaining useful life prediction of degrading systems subjected to imperfect maintenance: Application to draught fans



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ARTICLE INFO

Article history:

Received 12 December 2016

Received in revised form 20 June 2017

Accepted 9 August 2017

Available online 17 August 2017

Keywords:

Degradation modeling

First hitting time

Imperfect maintenance

Prognostics

Remaining useful life

ABSTRACT

Current degradation modeling and remaining useful life prediction studies share a common assumption that the degrading systems are not maintained or maintained perfectly (i.e., to an as-good-as new state). This paper concerns the issues of how to model the degradation process and predict the remaining useful life of degrading systems subjected to imperfect maintenance activities, which can restore the health condition of a degrading system to any degradation level between as-good-as new and as-bad-as old. Toward this end, a nonlinear model driven by Wiener process is first proposed to characterize the degradation trajectory of the degrading system subjected to imperfect maintenance, where negative jumps are incorporated to quantify the influence of imperfect maintenance activities on the system's degradation. Then, the probability density function of the remaining useful life is derived analytically by a space-scale transformation, i.e., transforming the constructed degradation model with negative jumps crossing a constant threshold level to a Wiener process model crossing a random threshold level. To implement the proposed method, unknown parameters in the degradation model are estimated by the maximum likelihood estimation method. Finally, the proposed degradation modeling and remaining useful life prediction method are applied to a practical case of draught fans belonging to a kind of mechanical systems from steel mills. The results reveal that, for a degrading system subjected to imperfect maintenance, our proposed method can obtain more accurate remaining useful life predictions than those of the benchmark model in literature.

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1. Introduction

The past decade has witnessed a growing interest in prognostics and health management (PHM) in both academic and industrial communities, such as electronics, energy industry, aeronautics and aerospace, manufacturing, automotive industry, nuclear plants, and fleet-industrial maintenance [1–7]. In general, PHM is an enabling discipline consisting of technologies and methods to assess the reliability of a product in its actual life cycle conditions to determine the advent of failure and mitigate system risk [8]. The key components of PHM include two aspects, i.e., prognostics and health management [1]. The former is usually dedicated to predicting the remaining useful life (RUL) of a system using the available health condition information, while the latter focuses on performing appropriate management activities (e.g., maintenance schedule and

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inventory control) to achieve the required safety and economic performance based on the obtained RUL predictions [9]. So far, RUL prediction has been well recognized as the key of PHM program due to its fundamental importance in PHM implementation [10–16].

In practice, the RUL of a system is defined as the length of time from the current time to the end of the useful life, and RUL prediction often refers to finding the mean estimate or the probability density function (PDF) of the RUL [15]. A large volume of studies have been dedicated to RUL prediction during these years; see [15] and [16] for comprehensive reviews. The RUL prediction methods have basically evolved upon two major paradigms: physics-based and data-driven. Due to the rapid development of sensor and condition monitoring (CM) techniques [17–20], the latter has received considerable attention in the literature [15]. For the importance of CM techniques, the readers are suggested to see [15–22]. Traditionally, the RUL of a system can be obtained on the basis of the time-to-failure data of a population of identical systems. However, it is usually costly or time-consuming, or even impossible to collect the time-to-failure data in reality [15]. As an alternative, the degradation-data-based methods have been extensively developed in the literature to infer the RUL of a degrading system [10,12,15,21]. Although the degradation-data-based RUL prediction methods have gained momentum over the last few years, it needs to be pointed out that almost all of the existing methods in the literature assume that the degrading systems are not maintained or are maintained perfectly (i.e., to an as-good-as new state) [15,16]. In such cases, the system's degradation is accumulated gradually during its usage, and the RUL can be derived accordingly [23]. However, in many realistic situations, the engineering systems may be maintained imperfectly (e.g., by adding lubrication, partial replacement, balance adjustment, etc.), periodically or aperiodically, over the lifecycle. Actually, imperfect maintenance can restore the degrading system to any health condition level between as-good-as-new (i.e., perfect maintenance) and as-bad-as-old (i.e., minimal maintenance) [24]. We note that maintenance activities can also accelerate the degradation of a system (named as worse maintenance in the literature) or even make the system fail or break down (named as worst maintenance in the literature), and also the health condition of a degrading system can be brought to a state better than the as-good-as new state because of technology improvement, but these situations do not belong to the imperfect maintenance activities defined in literature, and hence they are outside the scope of this paper. For a detailed classification and definition of the maintenance activities, please refer to [24–26], and the references therein.

Take the draught fan (belonging to a kind of mechanical systems) in the steel industry as an example; the draught fan plays a significant role in modern steel-making process due to its ability in dust elimination and gas recycling [27]. In practice, the vibration level of a draught fan is usually chosen as the CM data (i.e. degradation data) to characterize its health condition. To be specific, as the draught fan degrades, the vibration measurements increase gradually, mainly due to the mechanical wear of the rotor and the fouling on the impellers. As a result, once the measured vibration data reach the specified threshold level, the draught fan is announced to be failed, thereby being replaced by a new one. However, to improve the performance of the draught fan and prolong its useful life, balance adjustment techniques are performed aperiodically during its usage process. The basic principle of balance adjustment is to reduce the vibration level by cleaning the fouling on the impellers, lubricating the electromechanical components, etc. The balance adjustment process can bring the draught fan to a better health condition with a lower vibration level, but it cannot restore the performance of the draught fan to an as-good-as new state due to the existence of irreversible degradation mechanisms, e.g., the mechanical wear of the rotor. Thus, balance adjustment technique is just a kind of imperfect maintenance activities for draught fans. Predicting the RUL of draught fans operating like this, is an urgent reliability and safety demand of the steel-making process in steel industry.

For an engineering system subjected to imperfect maintenance, the system's failure progression process will be slowed down, and accordingly system life will be extended. Thus, a more generalized case in RUL prediction domain is to predict the RUL of degrading systems subjected to imperfect maintenance. Unfortunately, scarce literature can be found to address the degradation modeling and RUL prediction issue under such a scenario. The most relevant studies in literature took the influence of imperfect maintenance activities into account in scheduling condition-based maintenance strategies. Guo et al. [28] presented a maintenance optimization model for mission-oriented systems based on the Wiener process, where the system may experience imperfect maintenance activities. Le and Tan [29] developed an optimal inspection-maintenance policy for a multi-state system by taking imperfect maintenance and inspection into consideration. Do et al. [30] proposed a proactive condition-based maintenance strategy considering both perfect and imperfect maintenance activities, where the degradation trajectory between two successive maintenance activities were modeled by the gamma process, and the influence of the imperfect maintenance activity was described by a continuous random variable. Although the imperfect maintenance activity was involved in the aforementioned studies, they only focused on the maintenance schedule issue for degrading systems, and thus the influence of the imperfect maintenance activities was not considered in the RUL predictions. Based on the above survey, it is concluded that the problem considered in this paper has not been uniquely solved in literature. Thus, it can be of interest to develop a new model to predict the RUL of an engineering system subjected to imperfect maintenance. The challenges of predicting the RULs of the degrading systems subjected to imperfect maintenance activities may lie in the following two aspects: first, how to construct the degradation model of the degrading system subjected to imperfect maintenance activities, and second, how to derive the analytical PDF of the RUL when the influence of imperfect maintenance activities is considered.

In response to the above questions, we first model the degradation path by a nonlinear model driven by Wiener process with negative jumps, which are utilized to characterize the influence of the imperfect maintenance activities on the system's degradation. Specifically, 'negative' means that the influence of the imperfect maintenance is opposite to the degradation trend of the system, i.e., bringing the degrading system's health condition to a better state, and 'jump' means that imperfect

maintenance activity results in sudden improvement or change to the degradation of the system. By transforming the constructed nonlinear model with negative jumps crossing a constant threshold level to a Wiener process model, which is independent of the negative jumps, crossing a random threshold level, we obtain the PDF of the RUL with an analytical form. Based on the recorded degradation data and the maintenance archive, the unknown parameters in the degradation model are estimated via the maximum likelihood estimation (MLE) method. The proposed degradation modeling and RUL prediction method is finally applied to a case study of draught fans in steel mills. The main contributions of this paper are threefold. First, from the engineering point of view, we model the degradation path of a degrading system subjected to imperfect maintenance by a nonlinear model with negative jumps. Second, from the mathematical point of view, we derive the PDF of the first hitting time (FHT) of the nonlinear model with negative jumps analytically by a space-scale transformation strategy. Third, from the application point of view, we apply the proposed method to the real-world data of draught fans in steel industry.

The remainder of this paper is organized as follows. The problem description is given in Section 2. Section 3 specifies the derivation process of the PDF of the RUL. The parameter estimation procedure is presented in Section 4. In Section 5, a case study of draught fans is provided to demonstrate the effectiveness and superiority of the proposed method. Section 6 concludes this paper.

2. Problem description

2.1. A general degradation model for degrading systems subjected to imperfect maintenance

We describe the system's degradation evolution as a stochastic process, $\{X(t), t \geq 0\}$, with its realization $x(t)$. Specifically, for a degrading system subjected to imperfect maintenance, its degradation at time t should be a combination of the following two parts: the continuous smooth degradation at time t due to the normal degradation process and the accumulated effect of the imperfect maintenance activities until time t . Considering a practical system whose normal degradation increases gradually with its usage process, we can represent the system's degradation at time t by the following general model:

$$X(t) = X_c(t) - X_d(t), \quad (1)$$

where $X_c(t)$ represents the normal degradation at time t and $X_d(t)$ represents the accumulated influence of the imperfect maintenance activities on system's normal degradation until time t . Note that the minus sign before $X_d(t)$ means that the imperfect maintenance activities can bring the degrading system to a better health condition with a smaller degradation level than $X_c(t)$.

Based on the generalized degradation model (1), we provide a class of specific degradation models for RUL prediction below.

2.2. A Wiener process-based degradation model for degrading systems subjected to imperfect maintenance

If we use the Wiener process with a nonlinear drift to model the normal degradation part, $X_c(t)$, and a compound Poisson process to model the accumulated effect, $X_d(t)$, of the imperfect maintenance activities on the system's normal degradation, then the degradation at time t can be expressed by

$$X(t) = X(0) + \int_0^t \mu(\tau; \theta) d\tau - \sum_{j=0}^{N(t)} Y_j + \sigma_B B(t), \quad (2)$$

where $X(0)$ is the system's initial degradation, and without loss of generality, it is usually assumed to be zero in literature [31,32], $\mu(\tau; \theta)$ is the nonlinear drift coefficient with parameters θ , σ_B is the diffusion coefficient, and $B(t)$ is the standard Brownian motion (BM) [33]. BM with a drift has been recognized to be a favorable candidate for degradation modeling and RUL prediction, due to its mathematical advantages and physical interpretations [15], and until now, it has been widely adopted in a variety of industrial assets, such as, bearings [34], light-emitting diode lamps [35], hard disk drives [36], and gyroscopes [37]. The reason that we choose BM with a drift, rather than the other commonly-used stochastic process models (e.g., Gamma process, Inverse Gaussian process), to characterize the normal degradation of the system is that BM with a drift possesses a more flexible modeling structure, which can describe not only the monotonous degradation process but also the non-monotonous degradation process [1,5,15]. $N(t)$ is a Poisson process with rate λ characterizing the cumulative number of the imperfect maintenance activities up to time t (counting process) with $N(0) = 0$. Y_j (excluding $Y_0 = 0$) is an independent identically distributed random variable with PDF $f_Y(y; \vartheta)$ containing parameters ϑ , and it is used to characterize the influence of the j th imperfect maintenance on the system's normal degradation. Note that, $Y_0 = 0$ is obvious since $j = 0$ implies that the degrading system is not maintained at all. Y_j is also called intervention gain in the imperfect maintenance related literature [30].

Based on (1) and (2), it can be observed that $X_c(t) = X(0) + \int_0^t \mu(\tau; \theta) d\tau + \sigma_B B(t)$ and $X_d(t) = \sum_{j=0}^{N(t)} Y_j$. Then, we have the following remarks.

Remark 1. The nonlinear function $\int_0^t \mu(\tau; \theta) d\tau$ in $X_c(t)$ represents the accumulated effect of the system's normal degradation process [37–41]. Clearly, if the drift coefficient $\mu(\tau; \theta)$ is chosen to be a constant, then the normal degradation term $X_c(t)$ will reduce to the Wiener process with a linear drift in [28,32,42]. The reason that we choose a nonlinear function here is that, besides being more general than the linear model, nonlinearity exists extensively in practice [38,43,44].

Remark 2. If $Y_j = 0$ for all j , then (2) will reduce to the Wiener process with a nonlinear drift in [37–41].

Remark 3. As mentioned in the introduction, the health condition of the degrading system can be restored to any health condition level between as-good-as-new and as-bad-as-old after each imperfect maintenance activity. That is, imperfect maintenance activity can bring sudden improvement to the degradation of the system. The reason that we use compound Poisson process to characterize the influence of the imperfect maintenance activities lies in that compound Poisson process is typically appropriate to model the accumulation effect of sudden changes on system degradation [45–48].

Along with the above remarks, we can find that the degradation model in (2) possesses a general function form, which can subsume the commonly used Wiener process-based degradation models in literature as special cases.

2.3. RUL prediction for degrading systems subjected to imperfect maintenance

By employing the FHT concept, which has long been acknowledged as a mainstream definition of lifetime [23], the lifetime T (equivalently, FHT) of the degrading system (2) can be defined as

$$T = \inf\{t : X(t) \geq \omega | X(0) < \omega\}, \quad (3)$$

where ω is the threshold level, which can be determined by the well-accepted industrial standards, or the operational experiences [15,31,36,49].

Let $X(t_k)$ ($k \in N^+$) denote the degradation at time t_k ($0 \leq t_k < T$) over the system's life profile, and L_k (with its realization l_k) as the RUL at time t_k . Then, the following equation can be obtained from (2) based on the basic properties of BM and Poisson process:

$$U(l_k) = U(0) + \int_{t_k}^{l_k+t_k} \mu(\tau; \theta) d\tau - \sum_{j=0}^{N(l_k)} Y_j + \sigma_B B(l_k), \quad (4)$$

where $U(l_k) = X(l_k + t_k) - X(t_k)$ with $U(0) = 0$, and $N(l_k)$ represents a Poisson process with rate λ and $N(0) = 0$.

It can be found that (4) has a similar form with (2). Thus, the RUL L_k equals the FHT of $\{U(l_k), l_k \geq 0\}$ crossing the threshold $\omega - X(t_k)$. Hence, once the PDF of T is obtained from (3), the PDF of L_k can be derived accordingly.

3. Derivation of the PDF of the RUL

To derive the PDF of the RUL, we first derive the PDF of T below. Based on (1) and (2), we can obtain the counterpart of (3) as

$$T = \inf\{t : X_c(t) \geq \omega + X_d(t) | X(0) < \omega\}. \quad (5)$$

From (3) and (5), we can find that the derivation of the FHT of $X(t)$ crossing a constant threshold level ω can be transformed into the derivation of the FHT of X_c crossing a random threshold level $\omega_{th} = \omega + X_d(t)$.

Due to the existence of the nonlinear drift function, it is typically difficult to obtain the exact PDF of the FHT associated with (5). Thus, to get an approximated analytical PDF of the FHT, we will follow the method in [38] since the analytical result is more desirable in practice than the numerical simulations. The effectiveness of the analytical approximation method for the PDF of the FHT in [38] has been extensively validated by [38–41], which reveal that the derived PDF of the FHT is close to the exact one. Thus, such analytical method has been successfully applied in the RUL prediction practice, such as aluminum alloy [38], lithium-ion batteries [39], gyroscopes [37], fatigue cracks [41], and so on. Based on the above discussions, we have the following lemma for the presented model in this paper.

Lemma 1 [38]. Given threshold level ω_{th} , if $\mu(t; \theta)$ is a continuous function of time t in $[0, \infty)$, then the conditional PDF of the lifetime, T , defined in (5) can be formulated as

$$f_T(t | \omega_{th}) \cong \frac{\omega_{th} - \int_0^t \mu(\tau; \theta) d\tau + \mu(t; \theta)t}{\sqrt{2\pi\sigma_B^2 t^3}} \exp \left\{ -\frac{(\omega_{th} - \int_0^t \mu(\tau; \theta) d\tau)^2}{2\sigma_B^2 t} \right\}. \quad (6)$$

Eq. (6) is a conditional PDF of T on condition that the random threshold level ω_{th} is fixed. To obtain the unconditional PDF of T , the following result should be provided as the basis.

Theorem 1. Let $f_{T|N(t)=i, \bar{Y}_i=\bar{y}_i}(t|N(t)=i, \bar{Y}_i=\bar{y}_i)$ denote the PDF of T given $N(t)=i$ ($i \in N^+ \cup \{0\}$) and $\bar{Y}_i=\bar{y}_i$, then the following identity holds:

$$f_T^c(t|\omega_{th}) = f_{T|N(t)=i, \bar{Y}_i=\bar{y}_i}(t|N(t)=i, \bar{Y}_i=\bar{y}_i), \quad (7)$$

where $\bar{Y}_i$ (with its realization $\bar{y}_i$) is a summation random variable defined as $\bar{Y}_i \triangleq \sum_{j=0}^{N(t)} Y_j | N(t)=i$.

Theorem 1 is straightforward since $\omega_{th} = \omega + \sum_{j=0}^{N(t)} Y_j$. From (7), we can obtain the unconditional PDF of T by using the law of total probability as

$$f_T(t) = \sum_{i=0}^{\infty} \int_{\Omega_1} f_T^c(t|\omega_{th}) f_{\bar{Y}_i}^{(i)}(\bar{y}_i; \boldsymbol{\vartheta}) d\bar{y}_i P(N(t)=i), \quad (8)$$

where $f_{\bar{Y}_i}^{(i)}(\bar{y}_i; \boldsymbol{\vartheta})$ is the PDF of $\bar{Y}_i$ with parameters $\boldsymbol{\vartheta}$, Ω_1 is the parameter space of $\bar{Y}_i$.

Based on the results in (6)–(8) along with the Poisson process assumption in (2), we can obtain the following theorem.

Theorem 2. For a degrading system defined in (2), the PDF of the lifetime, T , can be formulated as

$$f_T(t) \cong \sum_{i=0}^{\infty} \int_{\Omega_1} \left\{ [(\omega + \bar{y}_i) - \int_0^t \mu(\tau; \boldsymbol{\theta}) d\tau + \mu(t; \boldsymbol{\theta})t] / \sqrt{2\pi\sigma_B^2 t^3} \right. \\ \left. \exp \left\{ -[(\omega + \bar{y}_i) - \int_0^t \mu(\tau; \boldsymbol{\theta}) d\tau]^2 / (2\sigma_B^2 t) \right\} f_{\bar{Y}_i}^{(i)}(\bar{y}_i; \boldsymbol{\vartheta}) \right\} d\bar{y}_i [\exp(-\lambda t)(\lambda t)^i] / i! \quad (9)$$

Based on **Theorem 2**, we further derive the PDF of the RUL, L_k . From (4), we can find that the RUL L_k is equal to the FHT of $\{U(l_k), l_k \geq 0\}$ crossing the threshold $\omega - X(t_k)$. Similar to the derivation process of **Theorem 2**, the following theorem can be obtained.

Theorem 3. For a degrading system defined in (2), the PDF of the RUL L_k at time t_k can be formulated as

$$f_{L_k}(l_k) \cong \sum_{m=0}^{\infty} \int_{\Omega_2} [\omega + \bar{y}_m - X(t_k) - \int_{t_k}^{l_k+t_k} \mu(\tau; \boldsymbol{\theta}) d\tau + \mu(l_k + t_k; \boldsymbol{\theta})l_k] / \sqrt{2\pi\sigma_B^2 l_k^3} \\ \exp \left\{ -(\omega + \bar{y}_m - X(t_k) - \int_{t_k}^{l_k+t_k} \mu(\tau; \boldsymbol{\theta}) d\tau)^2 / [2\sigma_B^2 l_k] \right\} f_{\bar{Y}_m}^{(m)}(\bar{y}_m; \boldsymbol{\vartheta}) d\bar{y}_m [\exp(-\lambda l_k)(\lambda l_k)^m] / m! \quad (10)$$

where $\bar{Y}_m \triangleq \sum_{j=0}^m Y_j$ ($m \in N^+ \cup \{0\}$) with its realization $\bar{y}_m$ and PDF $f_{\bar{Y}_m}^{(m)}(\bar{y}_m; \boldsymbol{\vartheta})$, and Ω_2 is the parameter space of $\bar{Y}_m$.

Eq. (10) is a general form for the PDF of L_k , and it can be applied for various problems with different distribution assumptions of Y_j and different choices of the drift coefficient function $\mu(\tau, \boldsymbol{\theta})$. The choosing of the distribution of Y_j and the function forms of $\mu(\tau, \boldsymbol{\theta})$ can be realized by engineering experiences, data fitting, etc. These choices can also be justified by statistical hypothesis testing methods in practice. In the following, we will choose two special distributions of Y_j as examples. For a specific application context, once appropriate function form of $\mu(\tau, \boldsymbol{\theta})$ is chosen, the PDF of L_k will be determined. Hereafter, these two special cases are also referred to as M_1 (i.e., Case 1) and M_2 (i.e., Case 2), respectively.

(1) Case 1: the influence of imperfect maintenance follows a normal distribution.

If the influence of each imperfect maintenance follows a normal distribution with $Y_j \sim N(\mu_Y, \sigma_Y^2)$, then $\bar{Y}_m \sim N(m\mu_Y, m\sigma_Y^2)$ from the property of normal distribution. In such a case, the following corollary can be obtained.

Corollary 1. If $Y_j \sim N(\mu_Y, \sigma_Y^2)$, then the PDF of RUL L_k at time t_k can be formulated as

$$f_{L_k}(l_k) \cong \sum_{m=0}^{\infty} \left(1 / \sqrt{2\pi l_k^2 (m\sigma_Y^2 + \sigma_B^2 l_k)} \right) \{ \omega - X(t_k) - v_k + \mu(l_k + t_k; \boldsymbol{\theta})l_k \\ + [m\sigma_Y^2(-\omega + X(t_k) + v_k) + m\mu_Y\sigma_B^2 l_k] / (m\sigma_Y^2 + \sigma_B^2 l_k) \} \\ \exp \left\{ -[\omega - X(t_k) - v_k + m\mu_Y]^2 / [2(m\sigma_Y^2 + \sigma_B^2 l_k)] \right\} [\exp(-\lambda l_k)(\lambda l_k)^m] / m! \quad (11)$$

where $v_k = \int_{t_k}^{l_k+t_k} \mu(\tau; \boldsymbol{\theta}) d\tau$.

The proof of Corollary 1 is put in the appendix.

(2) Case 2: the influence of imperfect maintenance follows a gamma distribution.

If the influence of each imperfect maintenance follows a gamma distribution with shape parameter, α , and scale parameter, β , i.e., $Y_j \sim Ga(\alpha, \beta)$, then $\bar{Y}_m \sim Ga(m\alpha, \beta)$ from the property of gamma distribution [33]. In such a case, we can obtain the following corollary.

Corollary 2. If $Y_j \sim Ga(\alpha, \beta)$, then the PDF of RUL L_k at time t_k can be formulated as

$$\begin{aligned} f_{L_k}(l_k) &\cong \left\{ [\omega - X(t_k) - v_k + \mu(l_k + t_k; \theta) l_k] / \sqrt{2\pi\sigma_B^2 l_k^3} \right\} \\ &\exp \left\{ -(\omega - X(t_k) - v_k)^2 / (2\sigma_B^2 l_k) \right\} \exp(-\lambda l_k) + [1 / (\beta^{m\alpha} \Gamma(m\alpha))] \\ &\sum_{m=1}^{\infty} \int_{\Omega_2} \left\{ [\omega + \bar{y}_m - X(t_k) - v_k + \mu(l_k + t_k; \theta) l_k] / \sqrt{2\pi\sigma_B^2 l_k^3} \right\} \exp(-\bar{y}_m / \beta) \\ &\bar{y}_m^{m\alpha-1} \exp \left\{ -(\omega + \bar{y}_m - X(t_k) - v_k)^2 / (2\sigma_B^2 l_k) \right\} d\bar{y}_m [\exp(-\lambda l_k) (\lambda l_k)^m] / m! \end{aligned} \quad (12)$$

This corollary can be obtained by substituting $f_{\bar{y}_m}^{(m)}(\bar{y}_m; \theta) = [\bar{y}_m^{m\alpha-1} \exp(-\bar{y}_m / \beta) I_{(\bar{y}_m \geq 0)}] / [\beta^{m\alpha} \Gamma(m\alpha)]$ into (10) directly, where $I_{(\bar{y}_m \geq 0)}$ is an indicator function with $I_{(\bar{y}_m \geq 0)} = 1$ if $\bar{y}_m \geq 0$, and $I_{(\bar{y}_m \geq 0)} = 0$ otherwise.

Although the PDF of the RUL has already been derived in this section, there still exist some unknown parameters. Next section will be devoted to estimating these unknown parameters.

4. Parameter estimation

We assume that: (1) we have the degradation data as well as the imperfect maintenance activities related data of N degrading systems operating under a similar context; (2) the data from different systems are statistically independent. Specifically, the following three classes of data are needed for parameter estimation. First, the data of each degrading system's total operating time and its total number of imperfect maintenance activities, denoted as $\{(T_n, p_n), n = 1, 2, \dots, N\}$, where T_n and p_n are, respectively, the total operating time and total number of imperfect maintenance activities of the n th system. Second, the data of the influence of imperfect maintenance activities, denoted as $\{y_{n,p}, p = 1, 2, \dots, p_n\}$, where $y_{n,p}$ represents the influence of the p th imperfect maintenance on the n th system at time $t_{n,p}$. In practice, $\{y_{n,p}\}$ can be obtained by recording the degradation improvement after each imperfect maintenance. Third, degradation data, denoted as $\{x_{n,d}, d = 1, 2, \dots, d_n\}$, where $x_{n,d}$ represents the d th degradation data of the n th system at time $t_{n,d}$, and d_n is the number of the degradation data of the n th system.

The unknown parameters in (2) can be classified into the following three parts: (1) the maintenance rate parameter, λ ; (2) the parameters ϑ in the PDF of Y_j , i.e., $f_Y(y; \vartheta)$. Such as, if $Y_j \stackrel{\text{iid}}{\sim} N(\mu_Y, \sigma_Y^2)$, then $\vartheta = (\mu_Y, \sigma_Y)$. Similarly, if $Y_j \stackrel{\text{iid}}{\sim} Ga(\alpha, \beta)$, then $\vartheta = (\alpha, \beta)$; and (3) the parameters in the normal degradation part $X_c(t)$, i.e., θ and σ_B . In summary, we denote all of the unknown parameters as $\Theta = (\lambda, \vartheta, \theta, \sigma_B)'$ and their estimated values as $\hat{\Theta} = (\hat{\lambda}, \hat{\vartheta}, \hat{\theta}, \hat{\sigma}_B)'$. These unknown parameters can be estimated by the aforesaid three kinds of recorded history data correspondingly in the following way.

4.1. Estimating λ

The estimation of λ can be realized according to the basic property of Poisson process. That is, if the counting process $N(t)$ for imperfect maintenance activities in (2) is a Poisson process with rate λ , then the times between two imperfect maintenance activities follows an independent identically distributed exponential distribution with parameter λ [33]. Accordingly, the likelihood function of the observed data $\{(T_n, p_n)\}$ with respect to λ can be formulated as

$$\ell_1(\lambda | \{(T_n, p_n)\}) = \prod_{n=1}^N \lambda^{p_n} e^{-\lambda T_n}. \quad (13)$$

Letting $\ell_1(\lambda | \{(T_n, p_n)\}) / d\lambda = 0$ in (13), we can readily obtain the estimated value of λ .

4.2. Estimating ϑ

ϑ can be estimated from $\{y_{n,p}\}$ via the MLE method based on the following likelihood function:

$$\ell_2(\vartheta | \{y_{n,p}\}) = \prod_{n=1}^N \prod_{p=1}^{p_n} f_Y(y_{n,p}; \vartheta). \quad (14)$$

The estimators $\hat{\vartheta}$ can be obtained by maximizing $\log \ell_2(\vartheta | \{y_{n,p}\})$ in terms of ϑ .

4.3. Estimating θ and σ_B

The parameters θ and σ_B can also be estimated by the MLE method based on the observed degradation data $\{x_{n,d}\}$. However, due to the existence of the imperfect maintenance activities related part $X_d(t)$ in (2), θ and σ_B cannot be estimated from $\{x_{n,d}\}$ directly. To facilitate the estimation of θ and σ_B , we have to transform the original degradation data $\{x_{n,d}\}$ based on (2) in advance.

Moving $X_d(t)$ in (2) from the right hand side to the left hand side yields

$$X(t) + \sum_{j=0}^{N(t)} Y_j = X(0) + \int_0^t \mu(\tau; \theta) d\tau + \sigma_B B(t). \quad (15)$$

Denoting the left hand side of (15) as $\Upsilon(t)$, we have

$$\Upsilon(t) = X(t) + \sum_{j=0}^{N(t)} Y_j, \quad (16)$$

where $\Upsilon(t)$ (with its realization $\gamma(t)$) can be regarded as an intermediate variable used for estimating θ and σ_B .

In such a case, the following counterpart relationship can be obtained relied on (15) and (16):

$$\Upsilon(t) = X(0) + \int_0^t \mu(\tau; \theta) d\tau + \sigma_B B(t). \quad (17)$$

Based on (15)–(17), θ and σ_B can be estimated by the following two steps: *Step 1*, transforming the observed degradation data $\{x_{n,d}\}$ to be $\{\gamma_{n,d}\}$ based on (16) and $\{y_{n,p}\}$; *Step 2*, estimating θ and σ_B based on (17) by using $\{\gamma_{n,d}\}$. Since $\{\gamma_{n,d}\}$ can be obtained directly from $\{x_{n,d}\}$ and $\{y_{n,p}\}$ based on (16), we lay our emphasis on the second step below. Note that, the term ‘degradation data’ in the following in this subsection refers to the transformed data, $\{\gamma_{n,d}\}$.

Considering the fact that the degradation data can only be observed at discrete time points in engineering, we denote the degradation observations of the n th system at two successive times $t_{n,d'}$ ($d' = 1, 2, \dots, d_n - 1$) and $t_{n,d'+1}$ as $\gamma(t_{n,d'})$ and $\gamma(t_{n,d'+1})$, respectively. Taking advantage of the independent increment property of the BM [33], we can formulate the likelihood function of all the degradation data with respect to θ and σ_B from (17) as

$$\begin{aligned} \ell_3(\theta, \sigma_B | \{\gamma_{n,d}\}) &= \prod_{n=1}^N \prod_{d'=1}^{d_n-1} \left(1 / \sqrt{2\pi\sigma_B^2(t_{n,d'+1} - t_{n,d'})} \right) \\ &\exp \left\{ -(\gamma(t_{n,d'+1}) - \gamma(t_{n,d'}) - \int_{t_{n,d'}}^{t_{n,d'+1}} \mu(\tau; \theta) d\tau)^2 / [2\sigma_B^2(t_{n,d'+1} - t_{n,d'})] \right\}. \end{aligned} \quad (18)$$

The $\hat{\theta}$ and $\hat{\sigma}_B$ can be obtained by maximizing $\log \ell_3(\theta, \sigma_B | \{\gamma_{n,d}\})$ in terms of θ and σ_B .

Substituting the estimated parameters $\hat{\Theta}$ into (9) and (10), and the associated corollaries, we can obtain the PDF of the lifetime, as well as the PDF of the RUL, for this kind of degrading systems with the influence of imperfect maintenance. In the forthcoming section, we will apply the obtained results to a practical case to validate the effectiveness and superiority.

5. Case study

This section is devoted to demonstrating the proposed RUL prediction method by the real-world data of the draught fans in steel mills. Specifically, we apply both M_1 and M_2 to the data. For comparison, the degradation modeling and RUL prediction method in [38] is cited as the benchmark model for this case study, and it is also termed as Si's approach hereafter. The difference between the Si's approach and our approaches is that the influence of the imperfect maintenance was not considered in the Si's approach, while it is considered in our approaches.

5.1. Problem description

As mentioned above, the draught fans are key devices in steel mills, especially for the pelletizing, sintering and smelting procedures. If the draught fans fail unexpectedly, the steel production along with the economic benefits of the steel mills will be greatly decreased. Even worse, the failure of the draught fans may lead to catastrophic accidents, thereby causing tremendous casualties and property loss. Hence, it is typically essential to predict the RUL of the draught fans during its usage process and take according decisions (e.g., scheduling replacement and ordering spare parts) in advance to avoid unexpected catastrophic failures. The degradation level of a draught fan is usually characterized by the vibration data collected from the sensors installed in the draught fan. The vibration of a draught fan might be caused by various kinds of factors. Based on the investigation and analysis of the deterioration mechanism of the draught fan operating in steel mill, the reasons that can cause the vibration of the draught fan mainly include rotor imbalance, rubbing, rotor misalignment, shaft bending, components loose, etc. In steel mills, the vibration data (amplitude) of the draught fans are recorded daily on a formal data sheet, and once the vibration signal reaches the specified threshold level, the draught fan will be replaced by a new one to maintain the normal steel-making process and avoid the catastrophic accidents. The threshold level of the draught fans in this case study is specified to be $\omega = 130$ (mm) according to the safety requirement of the steel mills. To improve the health condition of the convert fans, imperfect maintenance activities (i.e., balance adjustment techniques) are implemented aperiodically on the draught fans based on the experiences of the workers. But, once the vibration data is observed to reach the specified threshold level 130 (mm), it will be inappropriate to continue to perform imperfect maintenance activities, and thus the

failed draught fan should be replaced by a new one to ensure the operation safety and reliability. We collected a series of vibration data for three draught fans serving in the steel mills. These data include both the degradation data and the data associated with imperfect maintenance activities. Note that these three draught fans had served under a similar working condition. Below, we use two of them as training data and the left one as testing data, to validate the effectiveness and superiority of our proposed RUL prediction method. The degradation paths of these three draught fans are plotted in Fig. 1.

5.2. Results and discussions

5.2.1. Parameter estimation

The parameters in the degradation model need to be estimated for RUL prediction. As a precondition, the function form of $\mu(\tau; \theta)$ should be appointed first. According to the evolving trends of the observed degradation data of the draught fans, as shown in Fig. 1, together with the consultation with the engineers in the still mills, we choose $\mu(\tau; \theta)$ as a power function form, that is, $\mu(\tau; \theta) = a\tau^{b-1}$ with $\theta = (a, b)$. For a specific issue in practice, the function form of $\mu(\tau; \theta)$ could be chosen on the basis of the physical mechanism, expert knowledge, or the goodness of fit measures (e.g., AIC and BIC) [50]. Then, the unknown parameters needed to be estimated are $\Theta = (\mu_Y, \sigma_Y, \lambda, a, b, \sigma_B)'$ and $\Theta = (\alpha, \beta, \lambda, a, b, \sigma_B)'$, for M_1 and M_2 , respectively. Based on the recorded history data of the two training draught fans and the parameter estimation procedure specified in Section 4, the unknown parameters can be estimated, as shown in Table 1. In addition, to test the reasonableness of the assumptions used in the maintenance part (i.e., $X_d(t)$), we apply the statistical hypothesis testing methods to the collected maintenance related data in this case study. The following three assumptions are tested: (1) the Poisson process assumption for the arrival of the imperfect maintenance activities (equivalently to test that the length of time between consecutive imperfect maintenance activities follow an exponential distribution), (2) the normal distribution assumption for $f_Y(y; \vartheta)$, and (3) the gamma distribution assumption for $f_Y(y; \vartheta)$. The sample size is 8 in the Poisson process (or exponential distribution) test, and it is 6 in both the normal distribution test and gamma distribution test. The significance level is chosen to be 5% for all these three tests. Specifically, Lilliefors test is employed to test the exponential distribution and the normal distribution assumptions [51,52,53], and the Bootstrap based hypothesis testing method is used for the gamma distribution assumption [54]. The testing results of the recorded data all accept our null hypotheses. Thus, these assumptions used in our degradation model are all reasonable for this case study.

5.2.2. RUL prediction

Once the unknown parameters are estimated, we can predict the RUL of the testing draught fan based on the obtained results in Section 3. Specifically, we choose eight CM times at $t = 50, 100, 150, 200, 250, 300, 350, 400$ (days) for RUL prediction, which correspond to about the 10th, 20th, 30th, 40th, 50th, 60th, 70th, 80th degradation percentiles of the testing draught fan, respectively. The RUL predictions by M_1 and M_2 can be obtained by Corollary 1 and Corollary 2, respectively. We plot the predicted means and PDFs of the RULs at these chosen CM times by our approaches and the Si's approach in Fig. 2. Three observations can be drawn from Fig. 2. First, the predicted means of our approaches at the chosen CM times can track the actual RULs of the draught fan well (see Fig. 2(a) and (b)), but those predicted by the Si's approach, which does not

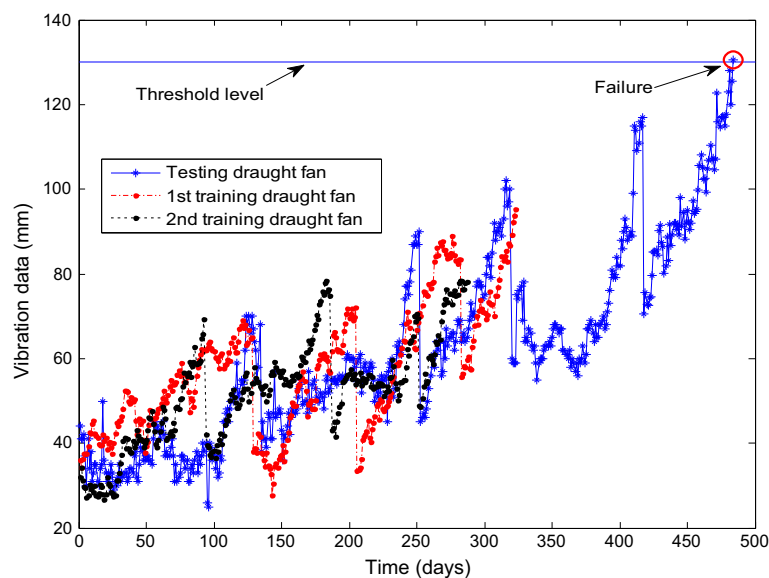


Fig. 1. The observed vibration data of the draught fans.

Table 1
The estimated parameters.

$\hat{a}$	$\hat{b}$	$\hat{\sigma}_B$	$\hat{\lambda}$
0.00015	2.3889	4.4135	0.0100
$\hat{\mu}_Y$	$\hat{\sigma}_Y$	$\hat{\alpha}$	$\hat{\beta}$
41.9980	6.4836	47.0414	0.9141

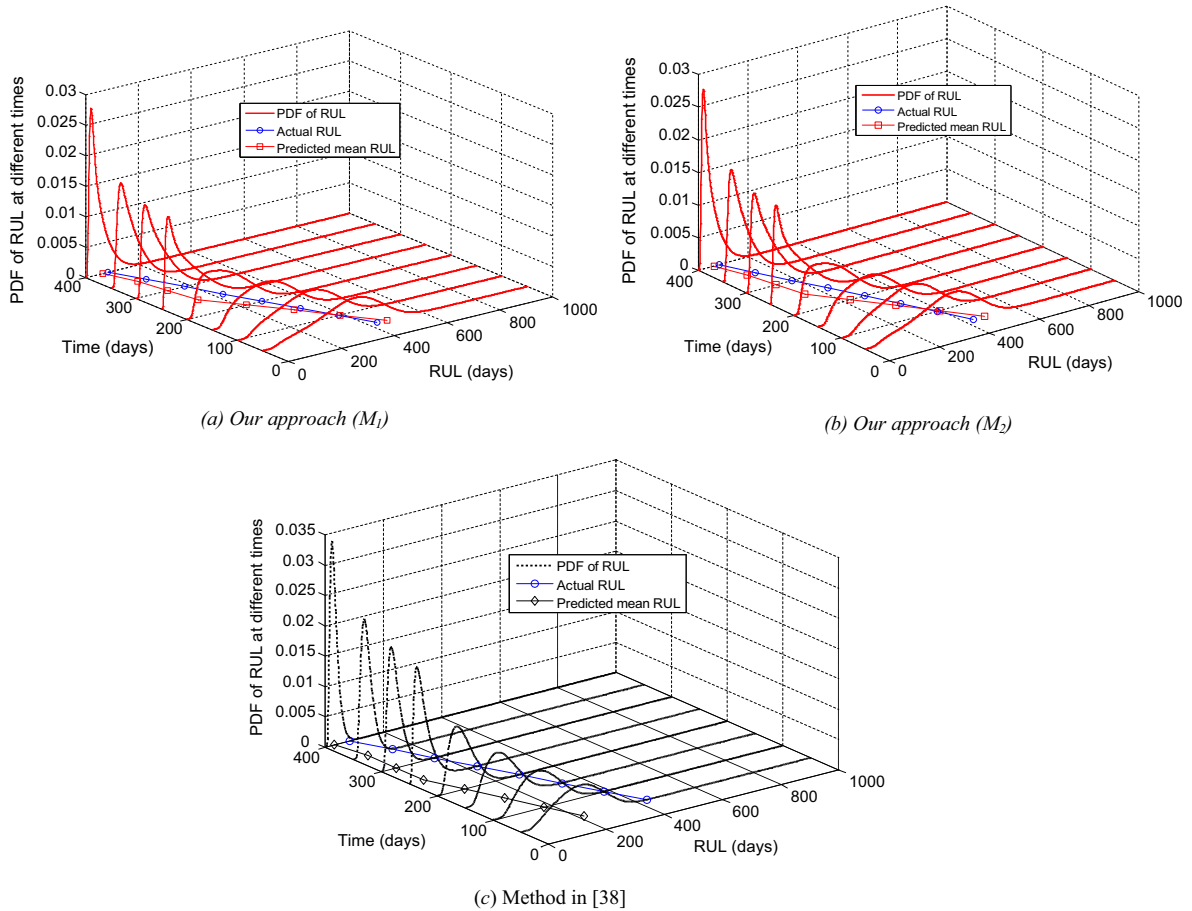


Fig. 2. Predicted RULs.

Table 2
The 95% CIs of the predicted rules (days).

CM time	Actual RUL	M_1	M_2	Si's approach
100	384	[76, 460]	[76, 467]	[61, 296]
200	284	[35, 325]	[35, 337]	[29, 185]
300	184	[19, 217]	[19, 239]	[17, 109]
400	84	[13, 141]	[13, 179]	[11, 67]

consider the influence of imperfect maintenance activities, lie far away from the actual RULs (see Fig. 2(c)). This implies that, compared with the Si's approach in literature (i.e., the benchmark model), our approaches can get more accurate RUL predictions. Second, the predicted PDFs of RULs by our approach can cover the actual RULs well, while most of the actual RULs are outside of the coverage of the predicted PDFs of RULs predicted by the Si's approach. This indicates that if the influence of imperfect maintenance activities is ignored in practice, one may get inappropriate RUL prediction results. Third, all of the predicted RULs by the Si's approach are much smaller than the actual RULs. This is because the Si's approach ignores the

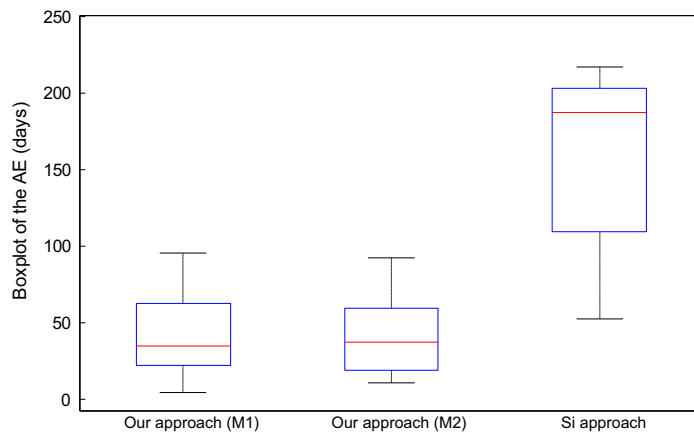


Fig. 3. Boxplot of the AE of the predicted mean RULs.

Table 3
The rul prediction errors.

	M_1	M_2	Si's approach
$MAPE_{RUL}$	20.19%	19.49%	63.46%
$RMSE_{RUL}$	50.5177	49.0285	168.5446
MAE_{RUL}	42.3179	41.6191	158.5029
CRA_{RUL}	0.7981	0.8051	0.3654

influence of imperfect maintenance activities, which can prolong the degrading system's lifetime. In other words, the RULs of the degrading system will be highly underestimated when using the benchmark model in literature.

In addition to the qualitative comparison in Fig. 2, we also provided the 95% confidence intervals (CIs) of the predicted RULs at four chosen CM times 100, 200, 300, and 400 (days) in Table 2. We can easily find from Table 2 that all the actual RULs at these four chosen CM times fall into the 95% CIs of the predicted RULs by our approaches M_1 and M_2 . However, the actual RULs at these four chosen CM times are all located outside of the predicted 95% CIs of the RULs by the Si's approach, and the actual RULs are all much higher than the upper bounds of the predicted 95% CIs by the Si's approach. These findings correspond to what we observed above from Fig. 2.

For a visual comparison of the prediction errors, we provide the boxplot of the Absolute Error (AE) of the predicted mean RULs by our approaches and the Si's approach in Fig. 3, where the AE at time t_q ($q = 1, 2, \dots, NM$) is defined as $AE_q = |RUL_q^A - RUL_q^P|$, where NM represents the number of chosen CM times for RUL predictions ($NM = 8$ in this case study), and RUL_q^A and RUL_q^P represent the actual and predicted RULs, respectively, at time t_q . From the boxplot, we can observe that both of our approaches have much smaller AEs than the Si's approach. To be specific, the medians and maxima of the boxplot by our approaches are, respectively, smaller than 50 (days) and smaller than 100 (days). However, the median and maximum of the boxplot by the Si's approach are nearly and more than 200 (days), respectively. This result implies that the predicted RULs by our approaches are much closer to the actual RULs than those predicted by the Si's approach.

To further compare the RUL prediction accuracies of our approaches and the benchmark approach in a quantitative manner, the following criteria, which are commonly used in the RUL prediction area [55], are employed: (1) Mean Absolute Percentage Error (MAPE) for RUL predictions, denoted as $MAPE_{RUL}$; (2) Root Mean Square Error (RMSE) for RUL predictions, denoted as $RMSE_{RUL}$; (3) Mean Absolute Error (MAE) for RUL predictions, denoted as MAE_{RUL} ; (4) Cumulative Relative Accuracy (CRA) for RUL predictions, denoted as CRA_{RUL} . Specifically, a smaller $MAPE_{RUL}$, or $RMSE_{RUL}$, or MAE_{RUL} corresponds to a higher prediction accuracy, but a larger CRA_{RUL} corresponds to a higher prediction accuracy. The calculated $MAPE_{RUL}$, $RMSE_{RUL}$, MAE_{RUL} , and CRA_{RUL} of our approaches and the benchmark approach among the chosen NM CM times are provided in Table 3. The following two observations can be obtained from Table 3. First, both of M_1 and M_2 clearly outperform the Si's approach in terms of all these four criteria. Second, though M_2 can decrease the RUL prediction errors slightly than M_1 , the prediction accuracies of M_1 and M_2 are very close, which implies that the choice of the distribution of Y_j , i.e., $f_Y(y; \vartheta)$, has tiny influence on the RUL predictions in this case study. These two results can also be observed from Table 2, Figs. 2 and 3.

In summary, three conclusions can be drawn from this case study. First, for the degrading system subjected to imperfect maintenance, our approaches can get more reasonable RUL predictions than the benchmark model in literature. Second, the predicted RULs of the degrading system will be highly underestimated by the benchmark model. Third, the choice of the function form of $f_Y(y; \vartheta)$ has little influence on the RUL prediction results in this case study.

6. Conclusions

This paper presented a general degradation modeling framework for incorporating the influence of imperfect maintenance activities into RUL predictions of a class of degrading systems. Specifically, to account for the influence of imperfect maintenance on RUL predictions, we first proposed a generalized degradation model based on a nonlinear-drift-driven Wiener process with negative jumps, which are used to characterize the influence of the imperfect maintenance activities, and then derived the approximately analytical PDF of the RUL based on the constructed degradation model, by a space-scale transformation scheme. The parameters in the degradation model were estimated based on the available degradation data and maintenance archive via the MLE method. We finally applied the proposed degradation modeling and RUL prediction method to real-life data of draught fans in steel mills to validate the effectiveness and applicability. Comparisons were also made with a widely adopted RUL prediction method in literature. We can conclude from the case study that the influence of the imperfect maintenance activities should be taken into account in degradation modeling and RUL prediction practice.

Successful implementation of the RUL prediction method proposed in this paper requires the engineers or users to focus on the following aspects: (1) to collect the historical degradation data, the maintenance-related data, and the operating time data; (2) to choose an appropriate degradation model and the distribution type characterizing the influence of imperfect maintenance based on domain knowledge or data fitting; (3) to estimate the unknown parameters based on the collected data. Finally, it is worth noting that although the proposed RUL prediction method was developed and validated by a real case for draught fans, it can also be applied to other degrading systems subjected to imperfect maintenance activities, or can be potentially applied to the other cases with a similar modeling framework, such as degradation process subjected to both smooth degradation and abrupt damages.

Acknowledgments

The work described in this paper is partially supported by NSFC under numbers 61603398, and 61573365, and the Young Talent fund of University Association for Science and Technology in Shaanxi, China.

Appendix A. Proof of Corollary 1

Proof. If $Y \sim N(\mu_Y, \sigma_Y^2)$, then the PDF of $\bar{Y}_m$ can be written as $f_{\bar{Y}_m}^{(m)}(\bar{y}_m) = \left(1/\sqrt{2\pi m\sigma_Y^2}\right) \exp\{-(\bar{y}_m - m\mu_Y)^2/(2m\sigma_Y^2)\}$. From the result in [Theorem 3](#), the following can be obtained with the help of [Lemma 2](#) in [\[39\]](#):

$$\begin{aligned} & \int_{\Omega_2} \frac{\omega + \bar{y}_m - X(t_k) - v_k + \mu(l_k + t_k; \theta)l_k}{\sqrt{2\pi\sigma_B^2 l_k^3}} \exp\left\{-\frac{(\omega + \bar{y}_m - X(t_k) - v_k)^2}{2\sigma_B^2 l_k}\right\} f_{\bar{Y}_m}^{(m)}(\bar{y}_m; \vartheta) d\bar{y}_m \\ &= E_{\bar{Y}_m} \left\{ \frac{\omega - X(t_k) - v_k + \mu(l_k + t_k; \theta)l_k + \bar{Y}_m}{\sqrt{2\pi\sigma_B^2 l_k^3}} \exp\left\{-\frac{(\omega - X(t_k) - v_k + \bar{Y}_m)^2}{2\sigma_B^2 l_k}\right\} \right\} \\ &= \frac{1}{\sqrt{2\pi l_k^2 (m\sigma_Y^2 + \sigma_B^2 l_k)}} \left\{ \omega - X(t_k) - v_k + \mu(l_k + t_k; \theta)l_k + \frac{m\sigma_Y^2[-\omega + X(t_k) + v_k] + m\mu_Y\sigma_B^2 l_k}{m\sigma_Y^2 + \sigma_B^2 l_k} \right\} \\ & \quad \times \exp\left\{-\frac{[\omega - X(t_k) - v_k + m\mu_Y]^2}{2(m\sigma_Y^2 + \sigma_B^2 l_k)}\right\} \end{aligned} \quad (19)$$

Substituting (19) into (10), we can derive (11) in [Corollary 1](#). $\square$

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